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Advanced Water Shut Off Application Using 2D Filler-Reinforced Polymer Gel Nanocomposite

A. Almohsin, EXPEC Research Center, Saudi Aramco, Dhahran, Saudi Arabia; M. R. Krishnan and E. H. Alsharaeh, Alfaisal University, Riyadh, Saudi Arabia; A. Busaleh, EXPEC Research Center, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

The current work reports the successful preparation of polymer gel-based nanocomposite using 2D-nanofillers for water shutoff applications. Polyacrylamide (PAM)-2D nanofiller composites were prepared by mixing a 4 wt% of aqueous PAM solution with a specific amount of organic cross-linkers of hydroquinone (HQ) and hexamethylenetetramine (HMT), potassium chloride (KCl) solution, and 2D-nanofillers. Subsequently, the PAM solution and sand particles were well mixed and cured at 150 °C for 8 hours to prepare the nanocomposite gels. The prepared samples were characterized using Fourier-transform infrared spectroscopy (FT-IR) and X-ray diffraction (XRD) for successful nanocomposite formation. The samples' equilibrium water absorption has reached as high as 90 g/g. Based on the results, the prepared PAM-2D-nanofiller composites can be potential candidates for water shutoff applications in the oil and gas industries.

Introduction

Though numerous methods have been reported in the literature, the systems based on crosslinked polymer gels are up-and-coming for water shutoff and water management. The significant advantages of the polymer gel systems include controlled gelation, high gel strength, and low cost. Most importantly, the polymer gels act as a high permeability layer, efficiently decreasing the heterogeneity of the formations, and therefore, the oil can be driven out. The polymer gel systems can be introduced in two ways: (i) injection of preformed polymer gel and (ii) injection of curable polymer solution at well conditions to the reservoir formation. However, the polymer gel system needs to improve its thermal, mechanical stabilities, and salinity resistance. Polymer gel systems are commonly employed in low-temperature-low-salinity wells (<80 °C and <50,000 mg/L). Meanwhile, there has been an increasing demand for polymer gel systems in high-temperature and high-salinity wells. Even after extensive research in this aspect, high-temperature, high-salinity wells still produce high water cuts. Therefore, it is essential to extend the application of polymer gels for high-temperature, high-salinity wells by overcoming their current limitations.

It is well-known that cross-linked polyacrylamide and their composite hydrogels were found to be potential candidates for water shutoff treatments. For instance, crosslinked PAM with certain metal ions such as Cr(III) and Al(III) are well-known for this purpose (Bai et al., 2015; El-Karsani et al., 2014; Sengupta et al., 2012). Recently, the utilization of organic crosslinkers to prepare the crosslinked PAM preparation has also been gaining attention because of their distinct advantages of controllable gelation kinetics and gel strength. Hydrolyzed PAM (HPAM) is a potential candidate among the different types of PAM because of its ability to crosslink with vast types of inorganic and organic crosslinkers. Also, the gelation mixture can be directly introduced to the reservoir formations so that the gelation will occur at the well conditions (high temperature and high pressure) (Sengupta et al., 2014). Various polymer systems, crosslinkers, and their respective gelation conditions were extensively reviewed and reported (Fathima et al., 2018). The crosslinked PAM composite gels with excellent water absorption capacities can act as potential plugging agents, reducing wellbore water production significantly. Generally, the PAM gels can absorb water up to 400 times its weight (Almohsin et al., 2019; Keishnan et al., 2020; Krishnan et al., 2019, 2022c).

Experimental Methods

Polyacrylamide (PAM) (low molecular weight PAM of M_w= 550,000 g/mol) was procured from Flotek. Organic crosslinkers such as HQ (99 % purity) and HMT (99 % purity) were purchased from Loba Chemie. The KCl was obtained from Scharlau (99 % purity). Saudi Aramco provided the macro sand (70/40 mesh). The polymer nanocomposite gels were prepared by a one-step in-situ method. The sand particles were coated with PAM hydrogel using an in-situ method. The typical experimental conditions were found in our previous studies. To prepare the PAM nanocomposite samples, a four wt.% PAM solution is thoroughly mixed with the organic crosslinker in 1:1 wt.% and KCl (2 wt.%) and 0.1 wt.% of 2D-nanofiller which is either commercial graphene (CG) or hexagonal boron nitride (h-BN). The weight percentages of all the chemicals concerning the amount of PAM used for the preparation. Afterward, a specific amount of PAM solution is mixed with sand particles and cured at high temperatures.

For instance, 5 g of sand particles is mixed with 5 g of PAM solution, which is well mixed using a mechanical stirrer and cured at 150°C for 8 hours. Hence, the PAM solution required to coat the sand was prepared by mixing a specified amount of PAM in D.I. water and dissolved for 1 hour. Then, the required amount of the organic crosslinkers and KCl were added and mixed further for another 15 min. Finally, the prepared PAM solution was mixed with the sand using a probe homogenizer. Then, the sand mixed with the PAM solution was cured. The Fourier transform infrared spectroscopy spectra were measured using FTIR and the Thermo Scientific Nicolet-iS10 instrument. The FT-IR spectra were measured from 4000 cm^{-1} to 650 cm^{-1} using ATR mode. The XRD patterns of the samples were measured using Rigaku MiniFlex 600. The X-rays were generated using Cu radiation [40 kV, 15 mA, Kα radiation (1.54 A°)], and the XRD patterns were recorded in the range 5-80°.

Results and Discussion

X-ray diffraction (XRD of PAM Nanocomposite Gels

XRD patterns of the samples were used to characterize the successful surface of the coating of PAM hydrogel onto the sand particles. Figure 1 shows the XRD patterns of the PAM nanocomposite gel samples compared to that of neat PAM, neat sand, and nanofillers of CG and BN. As evident from Figure 1, the neat sand is crystalline (quartz phase of SiO_2), and their characteristic peaks were detected at 21o, 26.5o, 42.7o, 46o, 50o, 60o, 64o, and 68°. PAM is found to be completely amorphous as it doesn't show any peak in XRD. At the same time, the nanofillers (CG and hBN) showed a characteristic peak at 26.5°. For the sand-PAM composite samples, the 011 peak of sand is observed to be largely decreased compared to that of neat sand. The absence of some of the original peaks or the decrease in their intensities in sand-PAM composite samples clearly indicates successful composite formation. Moreover, the original peaks of CG and hBN are observed to shift for the nanocomposite samples with nanofillers. The peak shifting can be correlated with the crosslinking of the nanofillers with PAM polymeric chains.

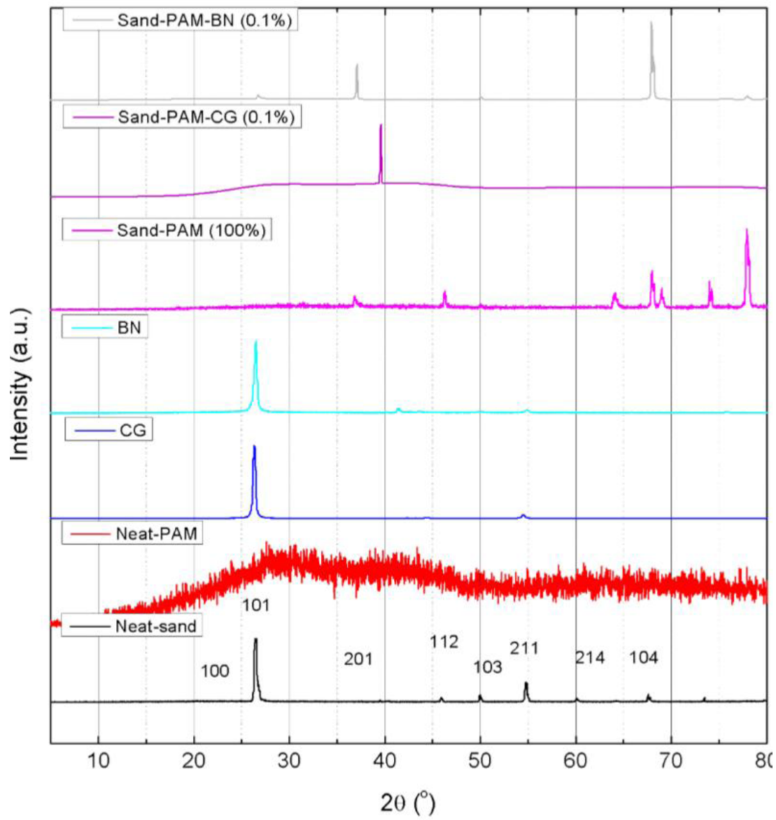

Figure 1—The XRD patterns of the PAM-coated sand in comparison to neat-sand and neat-PAM

Fourier Transform Infra-Red Spectra (FT-IR) of PAM Nanocomposite Gels

The chemical structure of the nanocomposite gels and the 2D-nanofiller crosslinking with PAM were studied using FT-IR. The FT-IR spectra of the PAM nanocomposite hydrogels compared to that of neat PAM, neat sand, and PAM hydrogels without 2D nanofillers are shown in Figure 2. The neat PAM samples exhibited characteristic peaks of PAM. The C=O bond was detected at 1642 cm-1, and the -OH group at 3382 cm^{-1}. Also, the neat showed its characteristic peaks at 1082 cm-1, 1175 cm-1, and 1632 cm-1, which are correlated to Si-O-Si, Si-O, and H-O-H stretching modes, respectively. The PAM composite with sand particles showed the characteristic peaks of sand and PAM molecules. In addition, the peak for the -OH group of the neat-PAM was identified at 3382 cm-1, but it was observed to shift to 3385 cm-1 for a sand-PAM composite sample, which is clear evidence of the sand-PAM composite formation. The nanocomposite samples with CG and BN showed their characteristic peaks in addition to new peaks due to the chemical crosslinking of the nanofillers (CG: 1430 cm^{-1}, 1590 cm^{-1}, and 1650 cm^{-1}; BN: 1430 cm^{-1}, 1550 cm^{-1}, and 1645 cm^{-1}) with PAM chains. The FT-IR spectra are clear evidence for successful nanocomposite gel formation. The FT-IR spectra complement the XRD results of the nanocomposite samples.

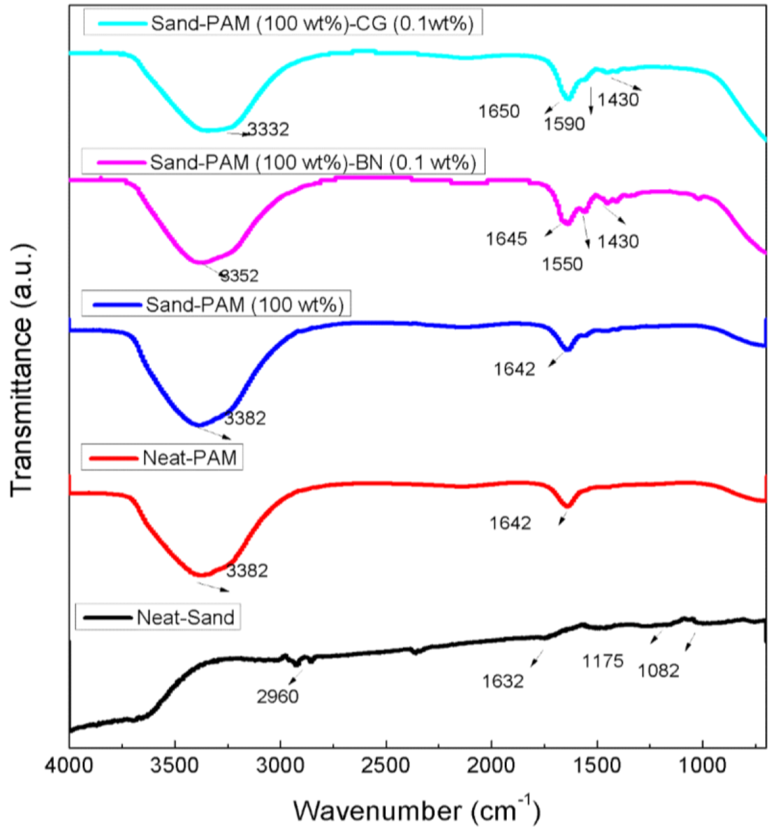

Figure 2—The ATR-FTIR spectrum of the PAM-coated sand and sand-PAM-CG and sand-PAM-BN in comparison to neat-sand and neat-PAM

Water Absorption Capacities of PAM Nanocomposite Gels

The water absorption properties of the PAM nanocomposite gels were evaluated using Equation 1. To measure the water absorption capacity, a specific amount of nanocomposite sample was immersed in excess water for several hours (for instance, 48 hours) to reach the equilibrium. The water absorption of the samples was calculated from the initial and final weights of the nanocomposite samples.

$$WaterAbsorptionCapacity(g/g) = \frac{w_2 - w_1}{w_1} \times 100 \quad (1)$$

W_2 (g) is the weight of the nanocomposite sample after the water absorption. W_1(g) is the weight of the nanocomposite sample before the water absorption process. Figure 3 shows the water absorption properties of the nanocomposite gels compared to those of neat PAM gel and those without the 2D nanofillers. As evident in Figure 3, the neat-PAM hydrogel showed the waster absorption at a maximum of 90 g/g, while the PAM-CG and PAM-hBN showed 80 g/g and 75 g/g, respectively. Interestingly, the incorporation of the 2D nanofillers into the PAM matrices resulted in a decrease in water absorption. The swelling ratio is as high as 90 g/g. This can be attributed to the networking or cross-linking of the 2D nanofillers with PAM hydrogels. The cross-linking of 2D nanofiller with PAM gel highly restricts the network expansion caused by the water absorption of the gel. Note that the higher the crosslinking density, the smaller the water absorption (the lower the network expansion is permitted) (Almohsin et al., 2023, 2022c, 2022a, 2022b; Krishnan et al., 2023c, 2023a, 2021c, 2021d; Michael et al., 2020a). These results are in line with XRD and FT-IR spectra—further incorporation of sand particles to the nanocomposite gels resulting in lower water absorption properties. For instance, sand-PAM composite hydrogels showed 20 g/g, while their respective nanocomposite hydrogels with CG and hBN showed 19 g/g and 18 g/g, respectively. Nevertheless, these absorption values are comparable. These results can be explained by the fact that the water absorption is directly proportional to the amount of PAM present in the gel, while the sand particles don’t absorb any water

(Krishnan et al., 2023b, 2022d, 2022b, 2021a, 2021b; Li et al., 2023, 2022, 2021; Michael et al., 2020b). The water absorption studies of the nanocomposite hydrogels suggested that the higher the amount of PAM, the higher the water absorption capacity due to the PAM's affinity towards the water molecules(Alsharaeh and Krishnan, 2020; Krishnan et al., 2022c; Krishnan and Alsharaeh, 2023).

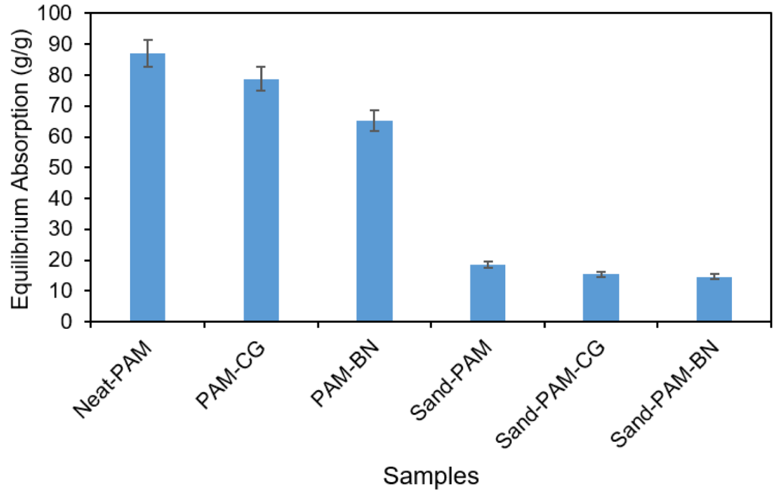

Figure 3—Water absorption of (a). Neat-PAM and PAM-2D nanofiller composites 48 hours of swelling in water.

Water Release Profile of PAM Nanocomposite Gels

Figure 4 shows the water release profile of the nanocomposite gel samples over 14 days. As evident from Figure 4, all the nanocomposite hydrogels can successfully retain the initially absorbed water molecules for up to four days. Afterward, the water molecules started to evaporate appreciably. On day fourteen, the samples were completely dried. More interestingly, the samples can be successfully applied to the next water absorption cycle and exhibit similar water absorption capacities. Similarly, the samples can be recycled up to 10 cycles of absorption-desorption processes with only a negligible decrease in the nanocomposite gels' water absorption capacities.

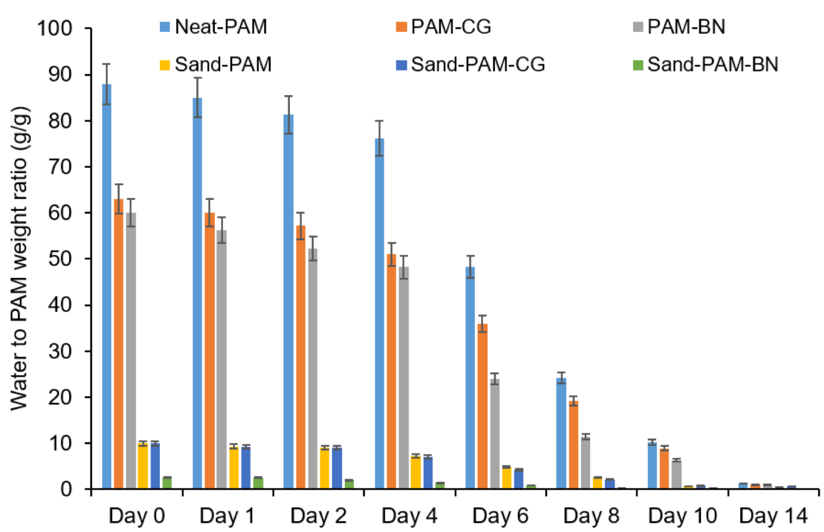

Figure 4—Water release profile of Neat-PAM and PAM-2D nanofiller composites for 14 days

Conclusion

The current work reports the facile preparation of sand-PAM-2D nanofiller composite hydrogels for water shutoff applications. The samples were prepared by mixing aqueous PAM solution (4 wt.%) with organic cross-linkers of HQ and HMT (1:1 wt.%), KCl solution, and 0.1 wt.% of 2D-nanofillers (CG or h-BN). Then, the above solution is mixed with specific sand and cured at 150 oC for 8 hours to get the nanocomposite gels. The nanocomposite gel samples were well characterized using FT-IR and XRD for the successful composite formation. The characterization results showed that the preparation of nanocomposite samples was successful. Compared to neat PAM, neat sand, and nanofillers, the nanocomposite sample preparation was confirmed with the respective peaks and peak intensities from FT-IR and XRD spectra. The FT-IR results showed new peaks that can be correlated to the mutual chemical interactions of the PAM nanocomposite hydrogel. XRD results showed that some of the original peaks disappeared for the polymer nanocomposite samples, which clearly indicates the successful PAM hydrogel matrix coating onto the sand surface. The PAM nanocomposite hydrogel showed an appreciable water absorption to a maximum of 90 g/g. This study also reports the effect of incorporating nanofillers into the PAM matrices on their water absorption characteristics. The PAM-2D nanofiller hydrogels showed a water absorption capacity comparable to the neat-PAM gels while significantly improving the mechanical integrity of the gel. Also, the water release profile of the nanocomposite gels was studied in detail. The prepared nanocomposite gel samples will have more significant potential for water shutoff applications.

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